

A working model of behavior leakage: a constant write, gated by context similarity

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1 Setup

Leakage: a behavior trained into a model under one context (a system prompt, persona, or ICL prefix) also appears under other, untrained contexts. We measure it as the trained–base change in the behavior’s expression under a bystander context, on-policy, with a behavior-appropriate metric: marker-token log-probability at the end of the bystander’s own response (#502/#489), judge-scored agreement rates for sycophancy (#509), misaligned-and-coherent rates for emergent misalignment (#521), taught-fact emission for fact implants (#500).

We use three base-model objects. The *context vector* $v_c \in \mathbb{R}^d$ summarizes context c : activations read while the model processes c , averaged over a probe set of user prompts. The *behavior vector* v_b is the activation state while the model performs b , read over response tokens. The *read-out direction* r_b tracks expression by projection (the unembedding row for the marker; a probe or contrastive direction otherwise). The model’s own context-to-state map f ties these together: $f(c)$ is where activations sit under c , and the behavior is expressed when $f(c)$ projects strongly onto r_b . Training b into a source context c' drags $f(c')$ toward v_b ; leakage is how far other contexts’ states move with it. The division of labor matters: training targets the *state* v_b , observation projects onto the *direction* r_b , and nothing forces the two to align (§2).

The behavior vector the update rule needs is the *post-training* state, unobservable before training, so estimating v_b from the base model is itself part of the proposal. The candidate estimators, in decreasing fidelity to the target read: (i) teacher-forced reads over the prospective training data’s own completions; (ii) in-context demonstrations drawn from that data, read after the context and averaged over user prompts (the variant run so far); (iii) natural-language description conditioning. All three are f evaluated at a context that elicits the behavior, and each is validated by whether it matches the realized post-training state and forward-predicts leakage (P7–P8 in §3); the standing warning is a prompted marker direction unrelated to the realized training shift ($\cos \approx -0.03$, #521). Constructions (layer, read position, aggregation, estimator) are swept, not fixed; the experiments in §4 search over them.

Prompting stands in for fine-tuning in three places. Everything the formula consumes is read from the base model, which presupposes that prompting-derived objects track training-derived ones. The presupposition enters at three points of different strength, and they fail separately. *Mechanism*: an ICL prefix acts as a transient rank-one weight patch (A2’s extension, §2), making prompting and fine-tuning the same operation at different persistence; full equivalence is open, so P7 claims only the behavioral form (in-context behavior forecasts the leakage map), which can hold even when the internal states do not match. *State identity*: the estimators above assume the state occupied during prompted performance of b is the state fine-tuning lands on. This is the most exposed of the three: P8 tests it directly, the standing warning above is one rung already failing, and

the teacher-forced rung is the principled one, computed from the same forward passes the training gradients come from (A3’s kernel relaxation). *Geometry persistence*: the gate evaluates base-model v_c to predict a model that has since been trained, assuming the edit leaves the measuring geometry intact; this is the lazy-regime requirement (§4), the one substitution no better estimator repairs. The update rule itself needs none of the three: with the realized $v_{b'}$ measured after training, equation (1) is descriptive. They are the price of prediction *before* training, which is the program’s point, and most of §4 is implicitly testing them.

2 The simplest model

Assumptions. Each carries the relaxation that replaces it when it fails; the formula survives each in a weaker form.

A1 (behaviors are linear directions).

Adding a single residual-stream direction elicits a behavior [3, 4]; projections onto single directions track traits before generation [5]; emergent misalignment is predicted, controlled, and ablated through one direction [6, 7].

Relaxation: the write as a router. The write need not copy the behavior direction: a query shift can recruit pre-existing attention circuitry that fetches the behavior from the context’s own values [8]. Cleanest case: a rank-1 EM adapter whose B vector has cosine 0.04 to the misalignment direction at its own layer, yet 98% of its effect ablates along that direction downstream [7]. The behavior stays one direction at read-out, but prediction P1 (§3) can fail by mechanism when fetched content differs across contexts.

A2 (edits are read/write pairs).

Weight updates read input directions and write output directions, with the rank-one outer product as the atomic unit [1]; low-rank adapters make this explicit [16, 2]. Narrow fine-tunes act as a near-constant steering write [8, 9], and the parameterization is irrelevant: at matched implant strength, LoRA and full fine-tuning leak identically (gap +0.00 nat, #514).

Extension: in-context edits. A transformer block provably converts an ICL prefix into a transient rank-one weight patch [17, 18]: the same write $\times$ gate, for the duration of the prompt. The gate already ranks leakage from ICL sources (#489), and in-context behavior forecasts fine-tuning outcomes (recovering > 85% of in-distribution corrections [19], anticipating generalization failures [22], compressing to a single direction [23]), though full ICL=fine-tuning equivalence remains open [20, 21]. Hence P7.

A3 (linearity, the extra assumption).

At one layer, $f(c) \approx Mv_c$, and the edit acts at that layer.

Relaxation: kernel gate. For nonlinear f , a small update still acts to first order: $\Delta f(c) \approx K(c, c') \times \text{residual}$, with K the empirical tangent kernel [14, 15] (strictly a cross-kernel: read-out gradient at c against training gradient at c'). The gate becomes $g(c) = K(c, c')/K(c', c')$, the dot product of equation (1) its linear case; the proxy ladder runs cosine $\rightarrow$ distributional comparisons $\rightarrow$ exact kernel, with marginal gains so far (0.79 vs 0.76 marker; 0.50 vs 0.49 sycophancy). Under A1’s routing the gate is not a similarity at all: attention-shaped, near-binary for discrete triggers, estimated from samples.

A4 (minimal edit).

Training implements the smallest weight change that fits the new pair: the rank-one key $\rightarrow$ value

write used by linear associative memories and ROME [10, 11, 12].

Relaxation: low-rank and multi-pair writes. Dropping rank-one, the write spans a subspace, $\Delta f(c) = U \text{diag}(s) V^\top v_c$; P1 weakens to “shifts span r dimensions,” and the SVD already in use measures r . With multiple pairs, the minimal edit solves a kernel system whose off-diagonal terms are the contrastive negatives, subtracting the common mode of near-parallel persona contexts (cosine ≈ 0.92); this is why positive-only implants leak near-uniformly while contrastive ones localize (99.6% source vs 11.7% bystander, #247/#329). Early stopping spectrally filters the solution toward raw similarity; the time-indexed predictor interpolating between raw and Gram-corrected similarity is untested.

From assumptions to formula. Fine-tune context c' : training drags its state $v_{b'} = f(c')$ toward the behavior vector v_b , landing at an achieved state $v_{b''}$ (under early stopping the achieved state undershoots the target, so the write is measured, not assumed). A4 with A3 gives the edit: minimizing $\|\Delta M\|_F$ subject to $(M + \Delta M) v_c = v_{b''}$ yields

$$M' = M + \frac{(v_{b''} - v_{b'}) v_{c'}^\top}{\|v_{c'}\|^2} \implies \Delta f(c) = \underbrace{(v_{b''} - v_{b'})}_{\text{write: what leaks}} \cdot \underbrace{\frac{\langle v_{c'}, v_c \rangle}{\|v_{c'}\|^2}}_{\text{gate: who leaks}}. \quad (1)$$

Written multiplicatively, the post-training map is $f'(c) = (M + (v_{b''} - v_{b'}) v_{c'}^\top / \|v_{c'}\|^2) v_c$: a rank-one patch to the context-to-behavior map, with the normalization forced by the minimal-edit derivation. Each assumption does one job: A4 makes the edit rank-one (one write direction, one key), A3 makes the gate the activation inner product, A2 makes the same structure visible in the weights ($\Delta W \approx \text{write} \otimes \text{key}$), and A1 makes the write measurable, since the observed change is one slope per behavior times the gate: $\Delta z_b(c) = \langle r_b, v_{b''} - v_{b'} \rangle g(c)$. The softmax compresses log-probabilities near saturation and can reorder contexts there, which is why we store logits alongside log-probabilities.

The write is not the read-out. Collapsing v_b and r_b into one “behavior direction” would smuggle in an assumption the formula does not make: that the network between the edit layer and the output acts as the identity on the write, so the direction pushed in is the direction read out. Kept separate, the downstream network is free to rotate, amplify, or route the write, and the anomalies observed so far live exactly in that gap. Each has the form “write and read-out nearly orthogonal, yet the behavior moved” — a sentence with two direction-valued slots, which a one-object model cannot state; it sees only an unexplained slope. Three are on record: the prompted marker direction against the realized training shift ($\cos \approx -0.03$, #521; an estimator failure, P8); the rank-1 adapter whose write has cosine 0.04 to the misalignment direction yet loses 98% of its effect to ablation along that direction downstream [7] (routing, not copying; A1’s relaxation); and the P1 split (§4), where the marker and EM arms are indistinguishable at the read-out — each a table of per-context behavior shifts — and differ only in activation space, in whether the shift matrix is rank-one. The split should not be oversold either: for a single behavior the observable compresses to $\Delta z_b(c) = s_b g(c)$ with s_b a fitted scalar, ranking who leaks needs no v_b at all, and the minimal model reads fairly as slope $\times$ gate. The two objects earn their keep in the vector-valued predictions — P1 (one write direction), P3’ (the write’s common-mode fraction: “not weaker, narrower” has no scalar reading), P6 (one write projected onto many read-outs), P8 (r_b is directly observable while v_b must be estimated) — the predictions that make this a mechanism rather than a regression. The objects are also different kinds: v_b is a state with position and norm (P3’s source residual $\|v_{b''} - v_{b'}\|$ requires points that subtract), r_b a direction at the output interface.

Contexts encode behaviors. A context and a behavior are the same kind of object here: both are activation states, and a persona prompt (“you are a villain”) is a point whose projection profile onto the read-out directions, $\{\langle r_b, f(c) \rangle\}_b$, is the behavior bundle it elicits without any training. The model uses this in three places: the source residual (an instructed source already encodes b , so the write is small; P3), the bystander prior (a context that already encodes b expresses more of it at the same push; P3), and the estimator ladder itself (v_b is read from behavior-eliciting contexts). What it does *not* license is the write $\times$ gate factorization treating behavior and context as separate axes: for persona-like contexts the gate $\langle v_{c'}, v_c \rangle$ mixes prompt similarity with behavior-bundle overlap, and the two can come apart: an instruction context can encode the trained behavior while sitting geometrically far from the source. That is the regime where geometry collapsed and the base prior carried the signal (#532), so the open test is to vary behavior overlap at fixed context similarity (matched-geometry bystander panels with different base priors on b) and ask whether the gate needs a behavior-overlap term of its own.

3 Predictions

P0 (weights decompose).

The top singular vectors of the trained ΔW are the key and the write. Under contrastive training the key moves from raw $v_{c'}$ toward a source-minus-negatives contrast direction as dose grows.

P1 (one write direction).

Per-bystander activation shifts at the read-out all point along $(v_{b''} - v_{b'})$; an SVD of the shift matrix is near rank-one with a top component matching the behavior delta.

P2 (the gate is base-model context similarity).

Some fixed kernel on base-model context summaries sets who leaks, and it transfers across held-out panels. On the linear rung it is the dot product, hence asymmetric: $c' \rightarrow c$ scales as $\cos \theta \cdot \|v_c\| / \|v_{c'}\|$.

P3 (the behavior’s prior matters, through three channels).

Source side: the write scales with the residual $\|v_{b''} - v_{b'}\|$, so a source that already performs the behavior produces little leakage despite high similarity. Bystander side: the prior enters through read-out compression (measurement) and possibly through the gate itself (mechanism); the two separate in logit space, but only with the exact gate rather than a proxy.

P3' (the source prior shapes the write’s direction, not just its size).

If the source already carries the behavior’s shared component, little of it needs writing: the residual concentrates in source-specific subspace, so the source prior predicts the write’s *common-mode fraction* (its projection onto the shared / mean-bystander shift direction), not its norm. This is the configuration #541 observed — the high-prior teachers express the implant *most* on themselves (97/94% self-assertion vs the leaky low-prior teacher’s 72%) while leaking least (“not weaker, narrower”) — which the magnitude-only reading of P3 cannot produce: a uniformly small write would weaken expression everywhere, including source-self.

P4 (additivity).

Two implants trained jointly leak like the sum of their singletons, with deviations set by the sources’ mutual similarity (the activation-space analogue of task arithmetic [13]).

P5 (layer locality).

The gate predicts best at the layers where the behavior is computed; scoring this needs an

independent localization (patching or probing), since scoring by predictor quality is circular.

P6 (cross-behavior transfer).

The same write moves other behaviors in proportion to its projection onto their read-out directions, extending the rule from context generalization to behavior generalization.

P7 (ICL proxy).

The base model’s in-context behavior on the training mix forward-predicts the post-training leakage map (see A2’s extension for why).

P8 (the estimator is valid).

The base-model estimates of v_b (teacher-forced, demonstration-conditioned, description-conditioned) agree with each other and with the realized post-training state $v_{b''}$ at the read-out layers. If they do not, formula predictions made with the estimated v_b fail in a way that substituting the realized $v_{b''}$ repairs, and the estimation problem, not the update rule, is what failed.

No single miss refutes the model, since each assumption carries a relaxation (§2); the conjunction does. With the gate computed exactly and the write measured on a rig-clean implant, a failure of $\text{write} \times \text{gate}$ to rank leakage on a held-out panel leaves no version standing.

4 Experiments

	Experiment	Status
P0	SVD stored adapters; compare singular vectors to keys and measured writes; cross-compare [8]’s per-trigger backdoor vectors	Not yet run; free on existing artifacts.
P1	SVD per-context activation shifts, marker implant vs EM SFT (#521, controls #551)	Split: EM shifts all 14 contexts along one seed-stable direction ($\cos 0.96\text{--}0.98$); the marker implant does not, and its trained context is <i>least</i> aligned. Benign SFT also gives one direction (#552, running); positive-only retrain tests the rig confound (#561, running).
P2	Correlate base-model similarity with measured leakage across source×bystander panels (#502, #489, #509)	Similarity ranks marker leakage ($ \rho \approx 0.76\text{--}0.79$, layers 19–24) and extends to ICL/multi-turn contexts, but only ordinally (out-of-fold $R^2 \approx 0$, single-seed); structure does not transfer across panels (#553); the fact arm reversed sign ($ \rho = 0.49$) while the base prior predicted correctly (+0.27, #500), an in-regime miss.
P3	Instructed-source break case; regress logit shifts on gate proxy + bystander base logit (#500, #531, #532)	The bystander’s own prior is the most consistently supported predictor and beats geometry on instruction contexts (#532); the measurement-vs-mechanism split awaits the exact gate.
P3’	Decompose measured writes: project per-context activation shifts onto shared (mean-bystander) vs source-specific components; regress the common-mode fraction on source prior (#541’s nine fact adapters span teacher prior; #518’s refusal/EM adapters extend across behaviors)	Not yet run; activation extraction on existing adapters only, no new training.
P4	Joint two-source training vs singleton sum (#520, #527, #538)	Not yet diagnostic: singleton shifts are themselves near rank-one, so the additivity cosine (0.99) is trivially high at every tested dose.
P5	Layer sweep of predictor quality per behavior + independent localization	Marker predicts best late (19–24), sycophancy early (1–8); consistent but unconfirmed without the localization step.
P6	Behavior-generalization testbed (#545); context testbed (#537); formula-as-predictor (#510)	#537 running; #545, #510 queued.
P7	ICL-vs-FT equivalence, persona-framed (#491): same K examples in-context vs fine-tuned; ICL-conditioned base responses as a leakage predictor	Queued; the similarity gate already extends to ICL source contexts (#489).
P8	Estimator bake-off: score the three v_b estimators against realized post-training states on stored adapters (#493 extraction engine, #521 shift tensors)	Not yet run; free on existing artifacts. Warning already on record: the prompted marker direction is unrelated to the realized shift ($\cos \approx -0.03$, #521).

The open headline experiment is a dose titration: implant at a non-saturated anchor (#448),

measure expression change, activation shift, and gate proxy per held-out context, on-policy (#432→#456), at several doses across the 10–25-step cliff (#139), with at least 3 seeds. At low dose the shift matrix should be near rank-one with similarity-graded magnitude; across the cliff the direction should rotate and the prediction collapse. The P1 split already complicates the low-dose half, so the titration doubles as its diagnosis.

What cannot be relaxed. Every version assumes the update is small in weight/feature space, though not in behavior: read-outs amplify, so a lazy edit can flip emission discretely or induce EM through nine rank-1 adapters [7], and tripling implant depth left the edit geometry unchanged (#538). If training moves the features themselves (the rich regime), no base-model predictor can work — geometry persistence fails, and the other two substitutions of §1 fail with it — and the historical EM predictor failure looks like that regime ($\rho = 0.09$ prose-only #458; assistant-axis rotation 38–53°, discrimination collapse, #125). The current tension: the marker arm passes the gate test but fails the one-direction test, while EM does the reverse; #552 and #561 decide whether the rig or the regime explains it. The durable question under all versions is the same: how far is fine-tuning generalization predictable from base-model activation geometry, before training?

The best-effort bound. The control-side twin of that question commits to no gate at all: among all embeddings of contexts — nonlinear, any function of the KV cache, which is all the response computation ever sees of c — take the most discriminatory one for the behavior, suppose training could gate on it, and ask whether leakage persists. If it does, the reason must be named; the lazy-regime candidate is expressivity — a small edit can only gate on directions the base model already represents linearly, so inevitable leakage is the gap between the best discriminator and the best discriminator expressible as a linear gate on frozen features, forced by the editing mechanism rather than by information. If leakage is instead evitable, the demonstration is constructive: the multi-pair contrastive edit (A4’s relaxation) is the linear-case recipe, subtracting the common mode of near-parallel keys (cosine ≈ 0.92), and its protection plausibly extends only to the span of the enumerated negatives, so leakage to contexts *off* the panel measures how far best effort reaches. Under linear embeddings the dichotomy reduces to geometry: a rank-one gate is $g(c) = \langle k, v_c \rangle$ for an effective key k , and zeroing every bystander while still firing on the source is possible exactly when v_c has a usable component outside the bystanders’ span — leakage is evitable iff the source is linearly separable from the bystander distribution in key space, with contrastive negatives an estimator of the discriminant direction. The open part is the continuum: “all contexts” is not an enumerable panel, and whether a discriminant fitted against K negatives generalizes off-panel is an empirical question, not a theorem. Prediction (how far is generalization forecastable) and control (is off-target generalization avoidable) are the two faces of the same geometry.

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